



CMPT 413/713: Natural Language Processing

Instruction tuning

Spring 2024
2024-03-11

Slides adapted from Anoop Sarkar

From LLMs to Helpful Assistants

How to build chatGPT from an LLM base model

<https://www.youtube.com/watch?v=bZQun8Y4L2A>

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw,
and sent them back to the earth so we could all see them.

<https://openai.com/research/instruction-following>

Training language models to follow instructions with human feedback

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OpenAI

<https://arxiv.org/abs/2203.02155>

GPT models (after GPT-3)



InstructGPT and GPT-3.5 [2022]

- Align responses to human feedback
- Instruction fine-tuning
- Reinforcement learning from human feedback
- Used in initial ChatGPT

- Supervised fine-tuning on human conversations
- Data where human will pretend to be user or AI assistant

GPT-4 [March 2023]

- Multimodal with images and text (GPT-4V)
- Larger, better model

- Human rank generated output
- Use reinforcement to improve generation

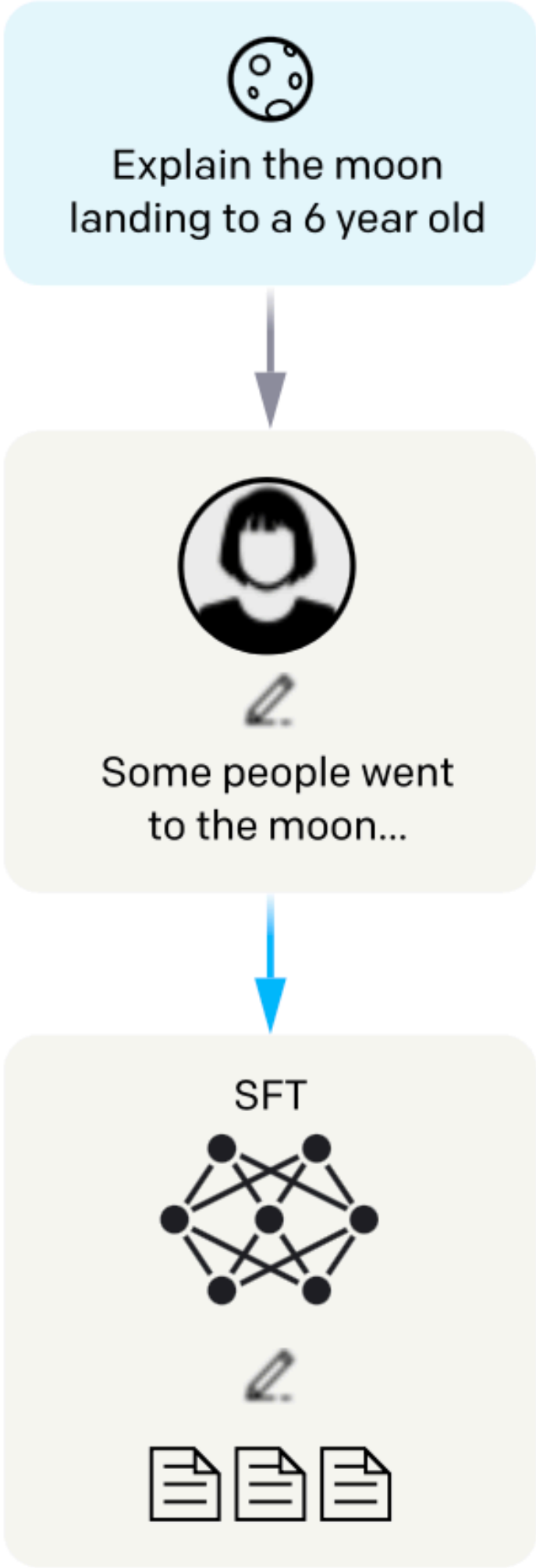
Step 1

Collect demonstration data,
and train a supervised policy.

A prompt is
sampled from our
prompt dataset.

A labeler
demonstrates the
desired output
behavior.

This data is used
to fine-tune GPT-3
with supervised
learning.



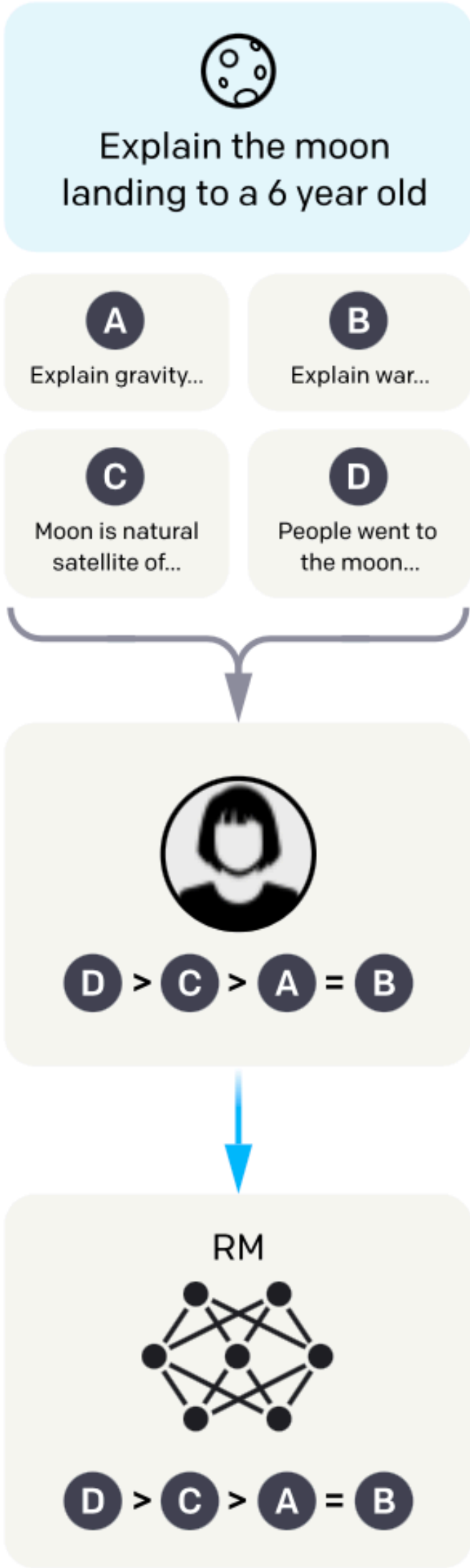
Step 2

Collect comparison data,
and train a reward model.

A prompt and
several model
outputs are
sampled.

A labeler ranks
the outputs from
best to worst.

This data is used
to train our
reward model.



Step 3

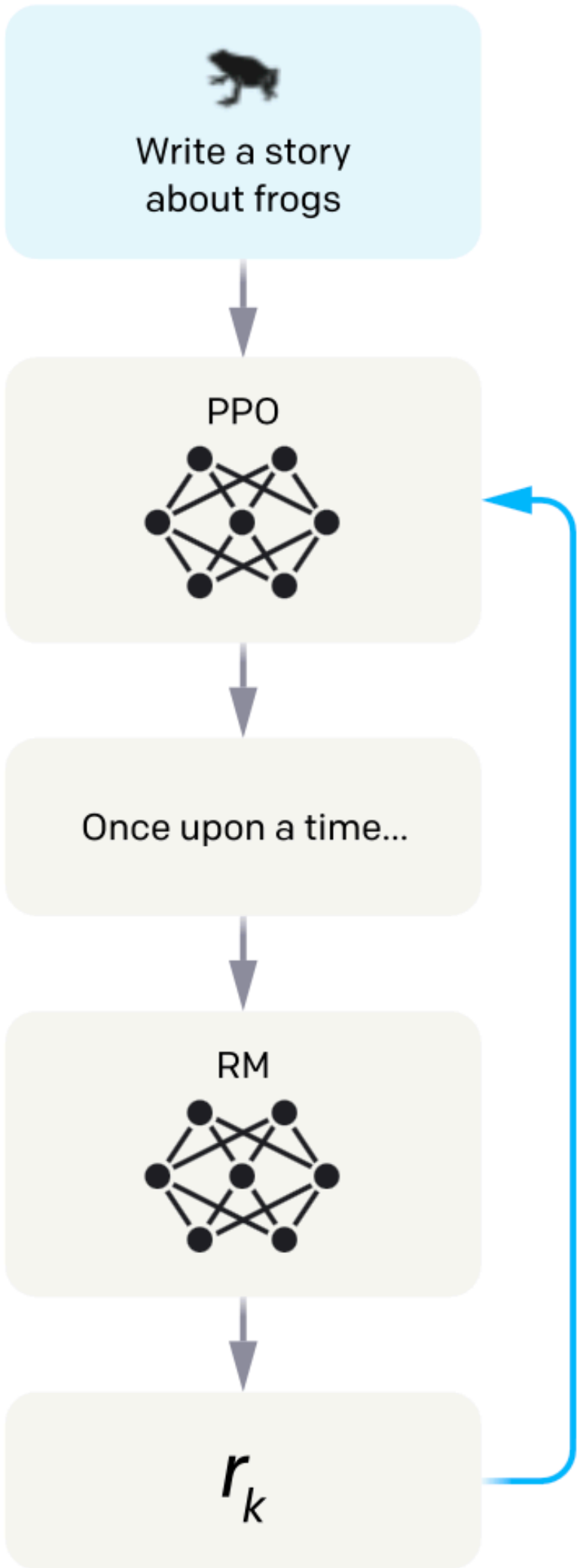
Optimize a policy against
the reward model using
reinforcement learning.

A new prompt
is sampled from
the dataset.

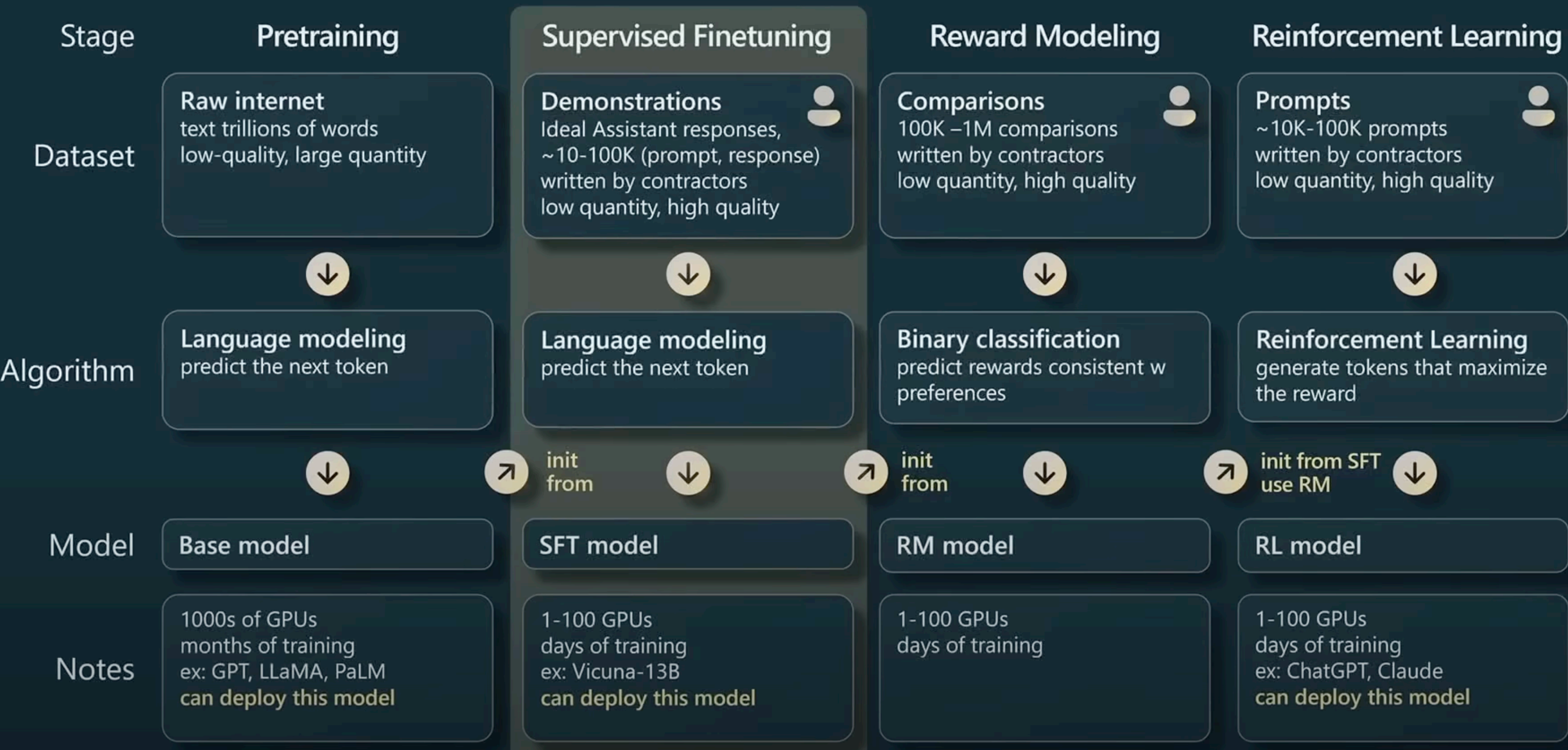
The policy
generates
an output.

The reward model
calculates a
reward for
the output.

The reward is
used to update
the policy
using PPO.



GPT Assistant training pipeline

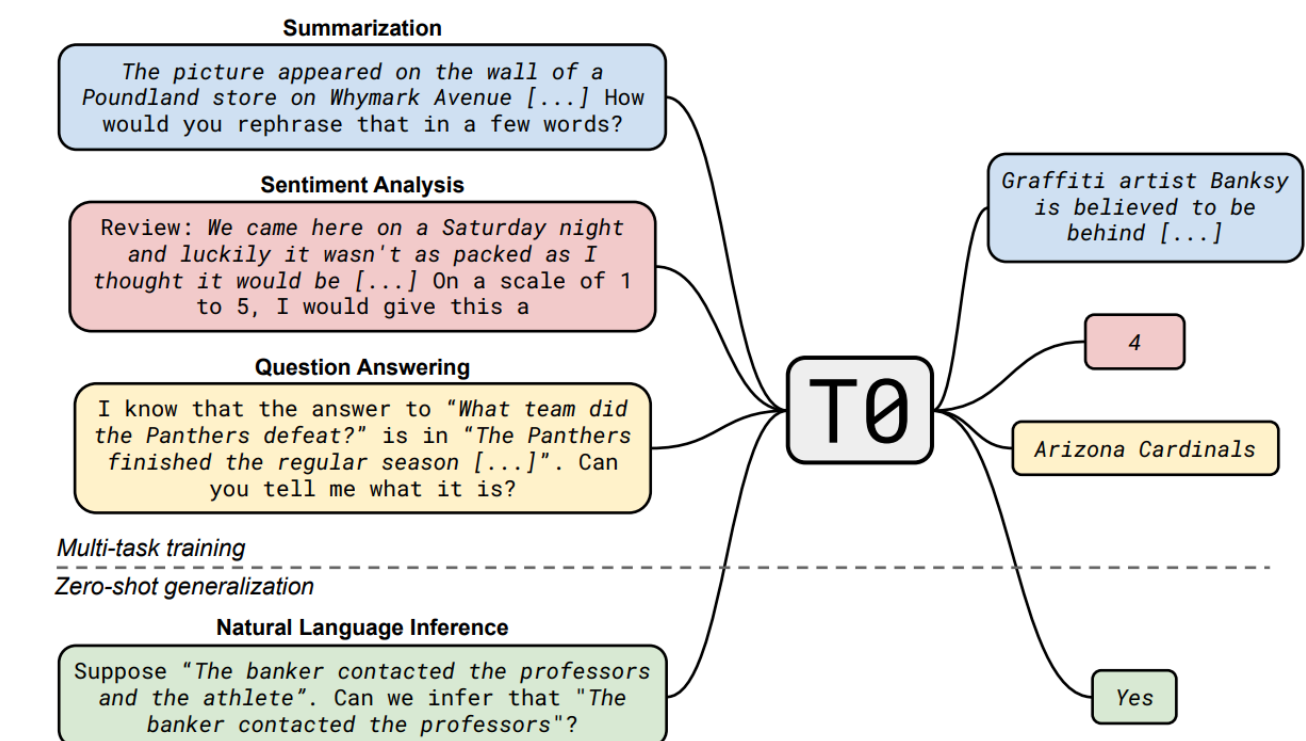


Supervised Fine-Tuning

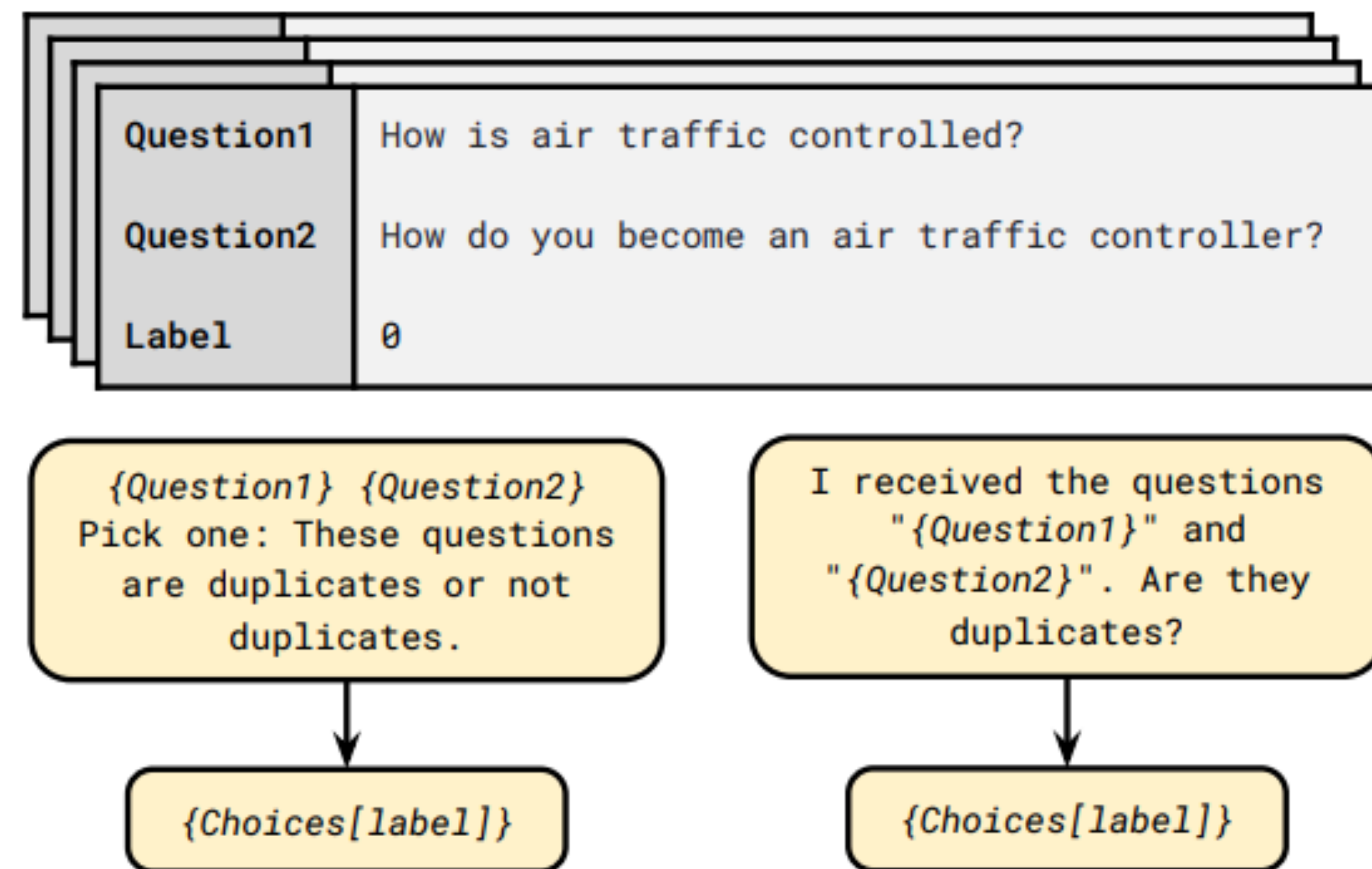
(instruction tuning without human data)

Instruction tuning

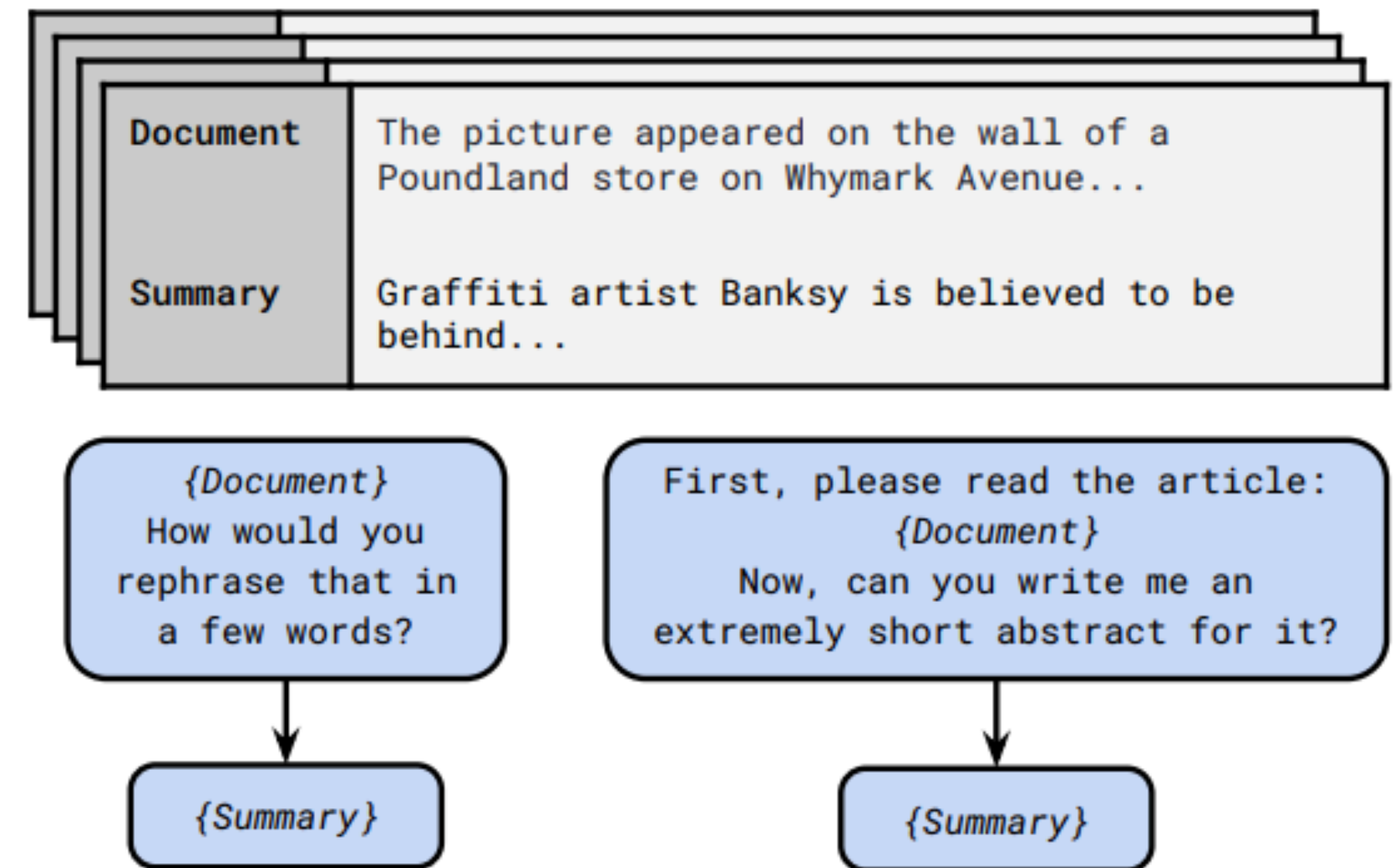
- Use templates to make them into instruction based dataset
- Text based format makes it natural for humans



QQP (Paraphrase)

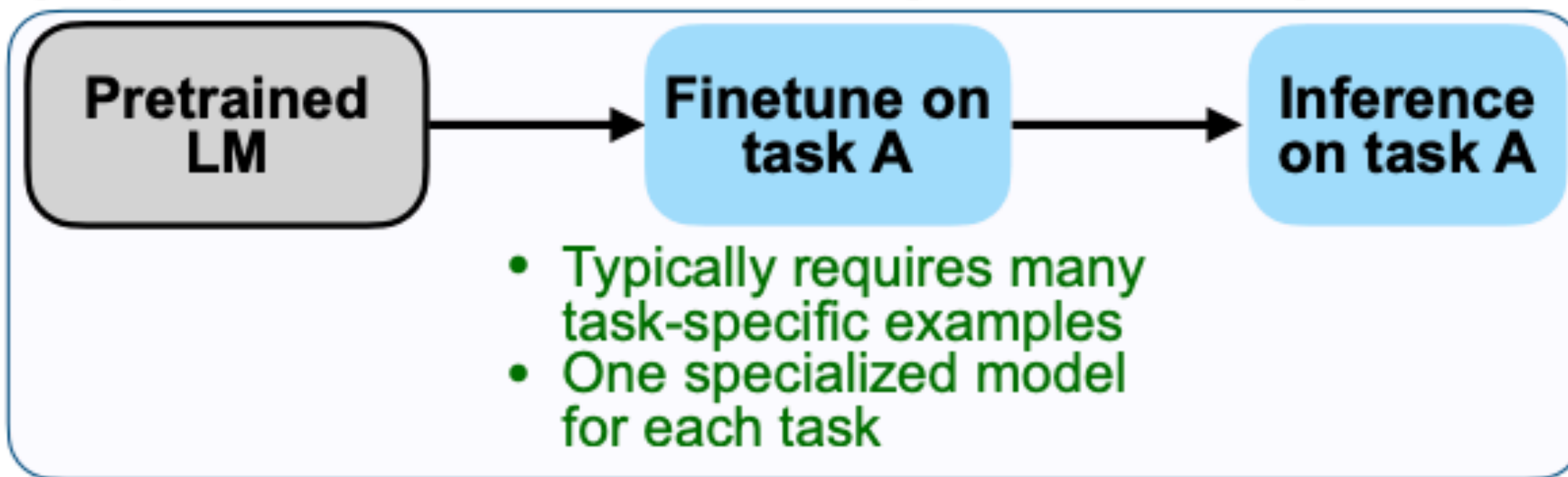


XSum (Summary)

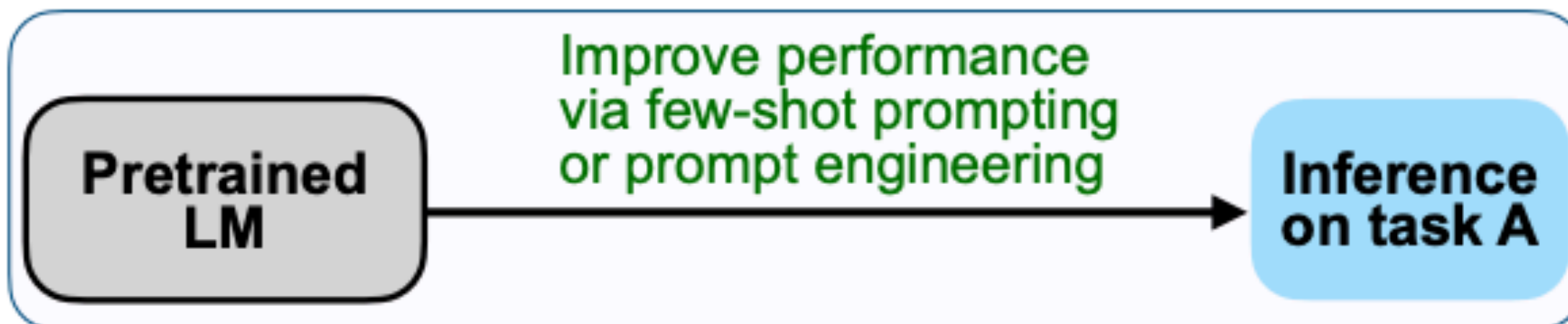


Instruction tuning

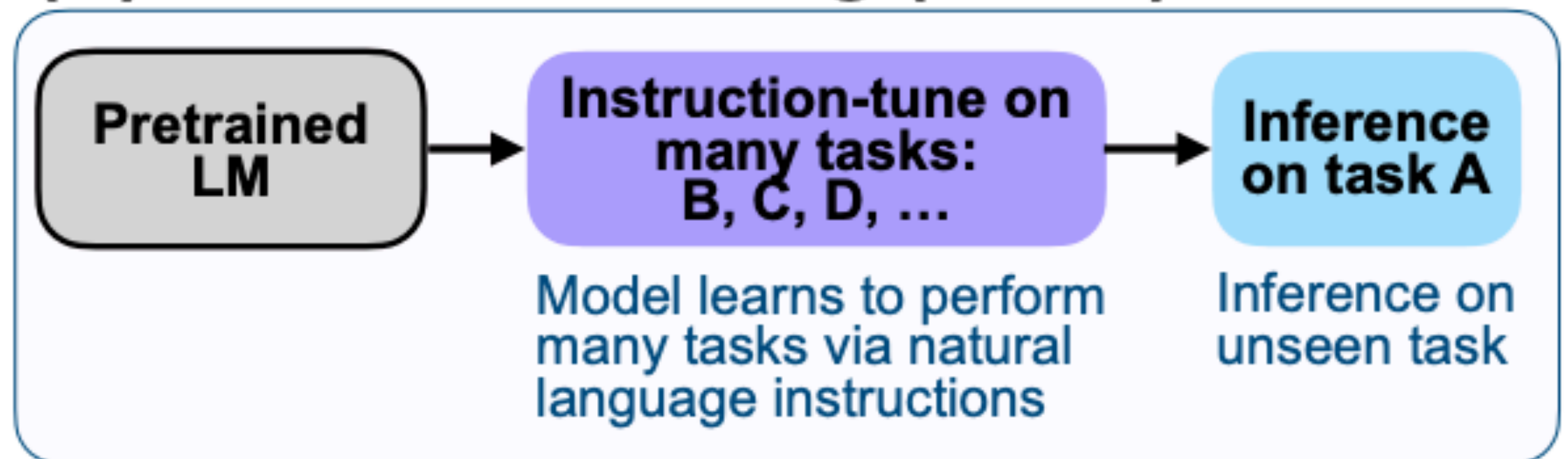
(A) Pretrain–finetune (BERT, T5)



(B) Prompting (GPT-3)

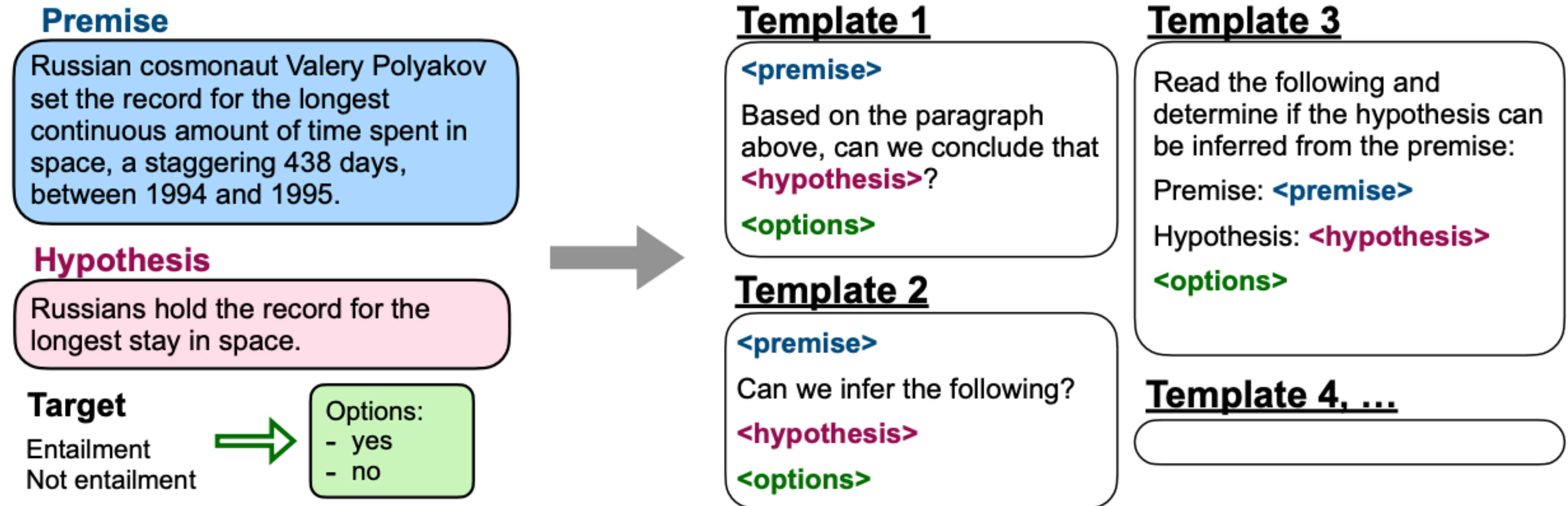


(C) Instruction tuning (FLAN)



Instruction tuning

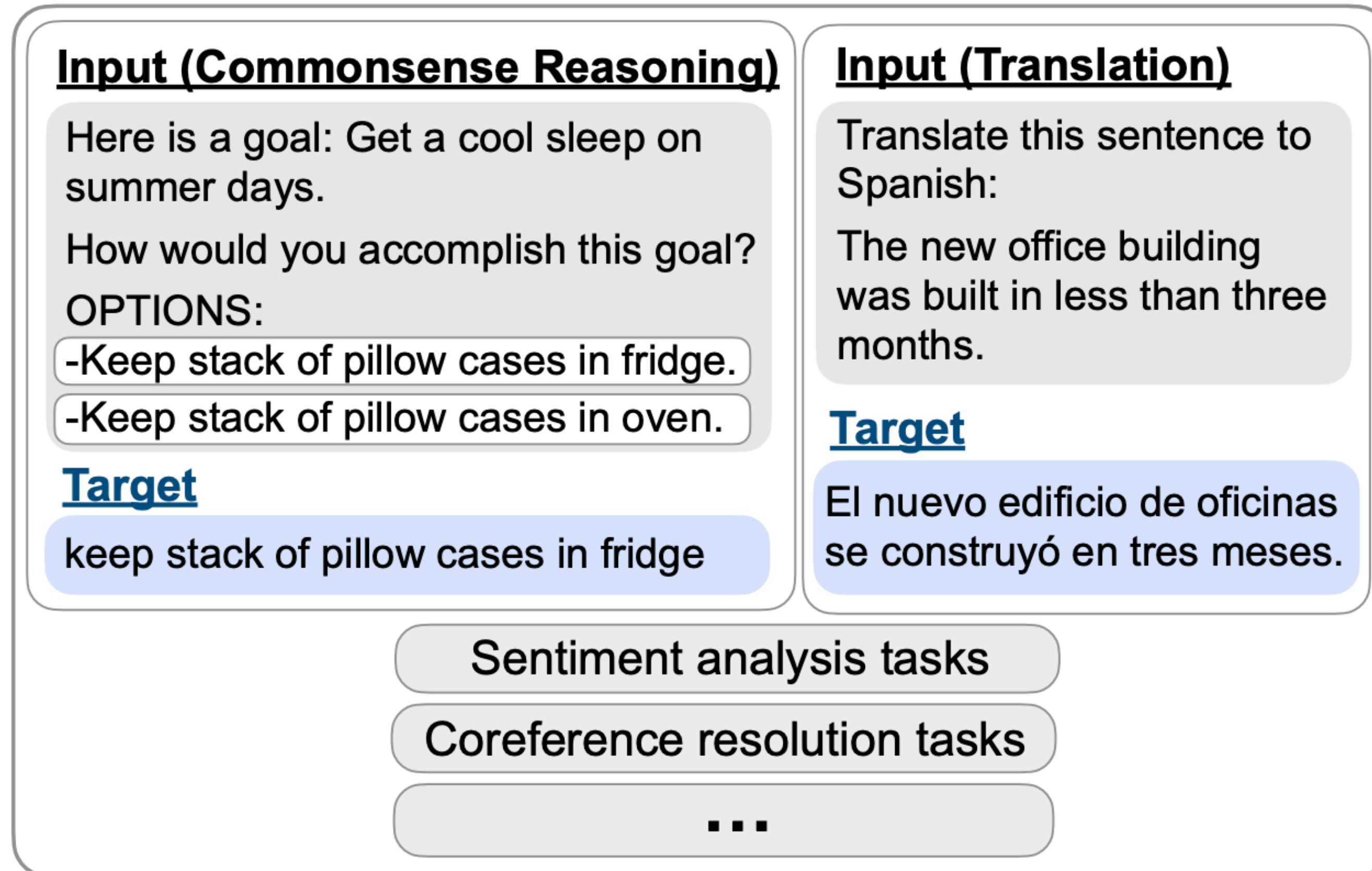
- Use templates to make them into instruction based dataset
- Text based format makes it natural for humans



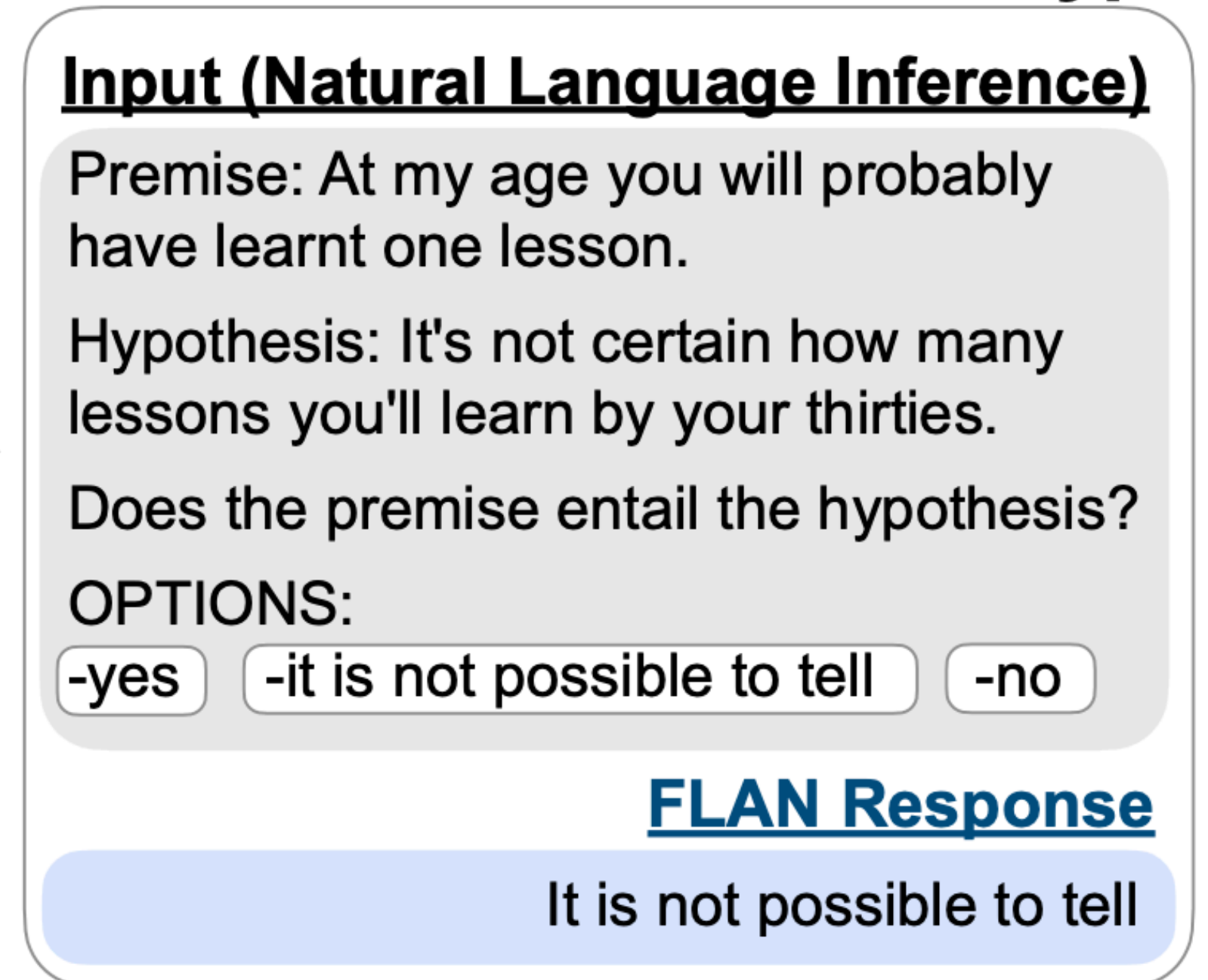
Instruction tuning

- Can be used on an unseen task type

Finetune on many tasks (“instruction-tuning”)

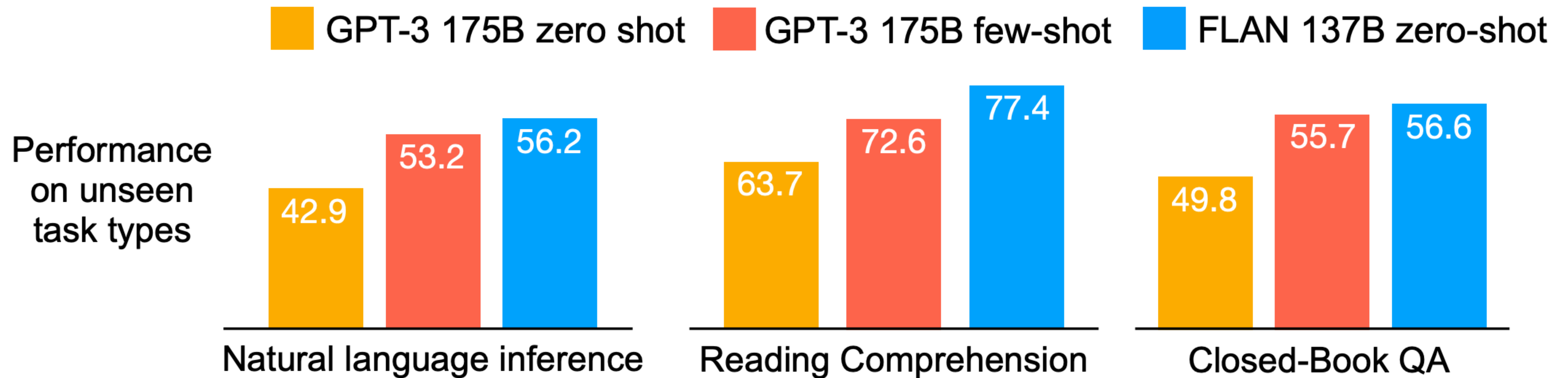


Inference on unseen task type



Instruction tuning

- Can be used on an unseen task type



Supervised Fine-Tuning

(instruction tuning with human data)

Prompt:

"Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research."

Response:

"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has little incentive to increase wages or provide better working conditions. Recent research has identified potential monopsonies in industries such as retail and fast food, where a few large companies control a significant portion of the market (Bivens & Mishel, 2013). In these industries, workers often face low wages, limited benefits, and reduced bargaining power, leading to a situation where they are dependent on the employer for their livelihood. This dependence can result in further suppression of wages and a decline in working conditions. Overall, the concept of monopsony is essential to understanding the dynamics of labor markets and the impact of market power on workers. Further research is needed to understand the extent and impact of monopsonies on the economy and to develop policies to address this issue. References: Bivens, J., & Mishel, L. (2013). The Pay of Corporate Executives and Financial Professionals as Evidence of Rents in Top 1 Percent Incomes. *Journal of Economic Perspectives*, 27(3), 57-78."

[example from OpenAssistant Conversations Dataset (OASST1)]

Table 1: Distribution of use case categories from our API prompt dataset.

Use-case	(%)
Generation	45.6%
Open QA	12.4%
Brainstorming	11.2%
Chat	8.4%
Rewrite	6.6%
Summarization	4.2%
Classification	3.5%
Other	3.5%
Closed QA	2.6%
Extract	1.9%

Table 2: Illustrative prompts from our API prompt dataset. These are fictional examples inspired by real usage—see more examples in Appendix [A.2.1](#).

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.
Rewrite	This is the summary of a Broadway play: "" {summary} "" This is the outline of the commercial for that play: ""

Excerpt of labeling instructions on the API prompt distribution

You are given a text-based description of a task, submitted by a user. This task description may be in the form of an explicit instruction (e.g. “Write a story about a wise frog.”). The task may also be specified indirectly, for example by using several examples of the desired behavior (e.g. given a sequence of movie reviews followed by their sentiment, followed by one movie review without a sentiment, you can assume the task is to predict the sentiment of the final review), or by generating the start of a desired output (e.g. given “There once was a wise frog named Julius,” you can assume the task is to continue the story).

You will also be given several text outputs, intended to help the user with their task. Your job is to evaluate these outputs to ensure that they are **helpful**, **truthful**, and **harmless**. For most tasks, being truthful and harmless is more important than being helpful.

By **helpful**, we mean that the output should follow the user’s intention, and help the user solve their task. Some examples of helpful behavior:

- Writing in clear language.
- Answering the question they meant to ask, even if they mis-asked it.
- Being sensitive to internationality (e.g. “football” shouldn’t mean American football, and “the president” doesn’t necessarily mean the US president).

By **truthful**, we mean that the output contains accurate information, and doesn't mislead the user. Some examples of truthful behavior:

- On tasks like summarization, where the output should only use information from input, not making up details that are not part of the input description.
- Not producing clearly false information about the world (e.g. making up facts or promoting conspiracies). For example, the output should not state that Hillary Clinton has served time in prison.
- Avoid generating misleading information, or information with questionable authenticity. For example, if the input asks "Why did Hillary Clinton go to jail?", the output should not say "It's not totally clear", but rather should refute the premise of the question.

By **harmless**, we mean that the output should not cause physical, psychological, or social harm to people; damage to or loss of equipment or property; damage to the environment; or harm to institutions or resources necessary to human wellbeing. Some examples of harmless behavior:

- Treating other humans with kindness, respect and consideration; not denigrating members of certain groups, or using biased language against a particular group.
- Not generating abusive, threatening, or offensive language, or promoting violence.
- Not writing sexual or violent content if it's not asked for.
- Not giving bad real-world advice, or promoting illegal activity.

Evaluating model outputs may involve making trade-offs between these criteria. These trade-offs will depend on the task. Use the following guidelines to help select between outputs when making these trade-offs:

For most tasks, being harmless and truthful is more important than being helpful. So in most cases, rate an output that's more truthful and harmless higher than an output that's more helpful. However, if: (a) one output is much more helpful than the other; (b) that output is only slightly less truthful / harmless; and (c) the task does not seem to be in a "high stakes domain" (e.g. loan applications, therapy, medical or legal advice, etc.); then rate the more helpful output higher. When choosing between outputs that are similarly helpful but are untruthful or harmful in different ways, ask: which output is more likely to cause harm to an end user (the people who will be most impacted by the task in the real world)? This output should be ranked lower. If this isn't clear from the task, then mark these outputs as tied.

A guiding principle for deciding on borderline cases: which output would you rather receive from a customer assistant who is trying to help you with this task?

Ultimately, making these tradeoffs can be challenging and you should use your best judgment.

Supervised Fine-tuning

- Data collected from human experts on Mechanical Turk or equivalent
- Detailed instructions are provided to obtain a high quality dataset
- Fine-tune GPT model on this data to maximize next token prediction loss

Reward Model Dataset

<https://github.com/openai/following-instructions-human-feedback>

Submit

Skip

«

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»

Total time: 05:39

Instruction

Summarize the following news article:

====
{article}
=====

Include output

Output A

summary1

Rating (1 = worst, 7 = best)

1

2

3

4

5

6

7

Fails to follow the correct instruction / task ? ☐ Yes ☐ No

Inappropriate for customer assistant ? ☐ Yes ☐ No

Contains sexual content ☐ Yes ☐ No

Contains violent content ☐ Yes ☐ No

Encourages or fails to discourage
violence/abuse/terrorism/self-harm ☐ Yes ☐ No

Denigrates a protected class ☐ Yes ☐ No

Gives harmful advice ? ☐ Yes ☐ No

Expresses moral judgment ☐ Yes ☐ No

Notes

(Optional) notes

Ranking outputs

To be ranked

B A team of researchers from Yale University and University of California, Davis studied the vocalization patterns of several different types of parrots. They found that parrots like to mimic human speech, and can produce a wide range of sounds, such as whistles, squawks, and other types of vocalizations...

Rank 1 (*best*)

A A research group in the United States has found that parrots can imitate human speech with ease, and some of them can even do so in the same way as humans. This group studied the sounds that parrots make in their natural habitats and found that they use their tongues and beaks in ways that are strikingly...

C Parrots have been found to have the ability to understand numbers. Researchers have found that parrots can understand numbers up to six. In a series of experiments, the parrots were able to identify the amount of food items under a number of cups...

Rank 2

E Scientists have found that green-winged parrots can tell the difference between two noises that are the same except for the order in which they are heard. This is important because green-winged parrots are known to imitate sounds. This research shows that they are able to understand the difference between sounds.

Rank 3

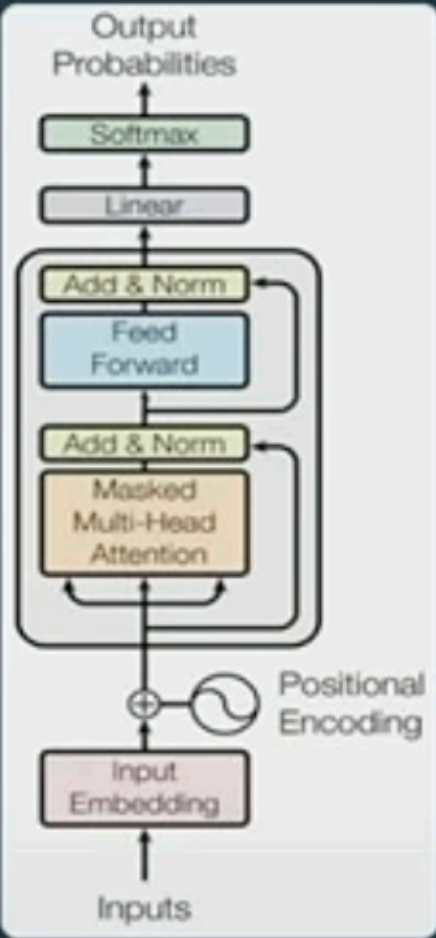
D Current research suggests that parrots see and hear things in a different way than humans do. While humans see a rainbow of colors, parrots only see shades of red and green. Parrots can also see ultraviolet light, which is invisible to humans. Many birds have this ability to see ultraviolet light, an ability

Rank 4

Rank 5 (*worst*)

Reward Model Training

Blue are the prompt tokens, identical across rows
Yellow are completion tokens, different in each row
Green is the special <|reward|> token "readout"
Only the outputs at the green cells is used, the rest are ignored



loss function measures the predicted rewards' consistency with the labeled ordering

B↓	T→									
	prompt	...	...	completion 1	...	...	< reward >			
	prompt	...	...	completion 2	...	...	...	...	...	< reward >
	prompt	...	...	completion 3	...	< reward >				

Reward Model Training

- Let θ be the parameters for the <reward> token which is appended at the end of each completion
- Data: Prompt | Completion | <reward>
- K is the number of responses ranked by humans ($K=\{4,9\}$). D is the dataset of human comparisons
- This produces $\binom{K}{2}$ comparisons for each prompt
- Loss function:
$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l) \sim D} [\log(\sigma(r_\theta(x, y_w) - r_\theta(x, y_l)))]$$
- $r_\theta(x, y)$ is the scalar reward for prompt x and completion y . y_w is preferred to y_l
- Train all $\binom{K}{2}$ comparisons in a single batch.
- Training the 175B model does not work, instead fine-tune a smaller 6B model to predict reward.

Bradley-Terry ranking

- The BT model is a probability model for the outcome of pairwise comparisons.
- Given a pair of individual responses i and j
- The probability of preferring $i > j$ is given by
- $$P(i > j) = \frac{p_i}{p_i + p_j}$$
- The Bradley–Terry model can be used in the forward direction to predict outcomes,
- But is more commonly used in reverse to infer the scores p_i given an observed set of outcomes (preferences from humans)
- More general models exist: e.g. Plackett-Luce models (but not used for RLHF)

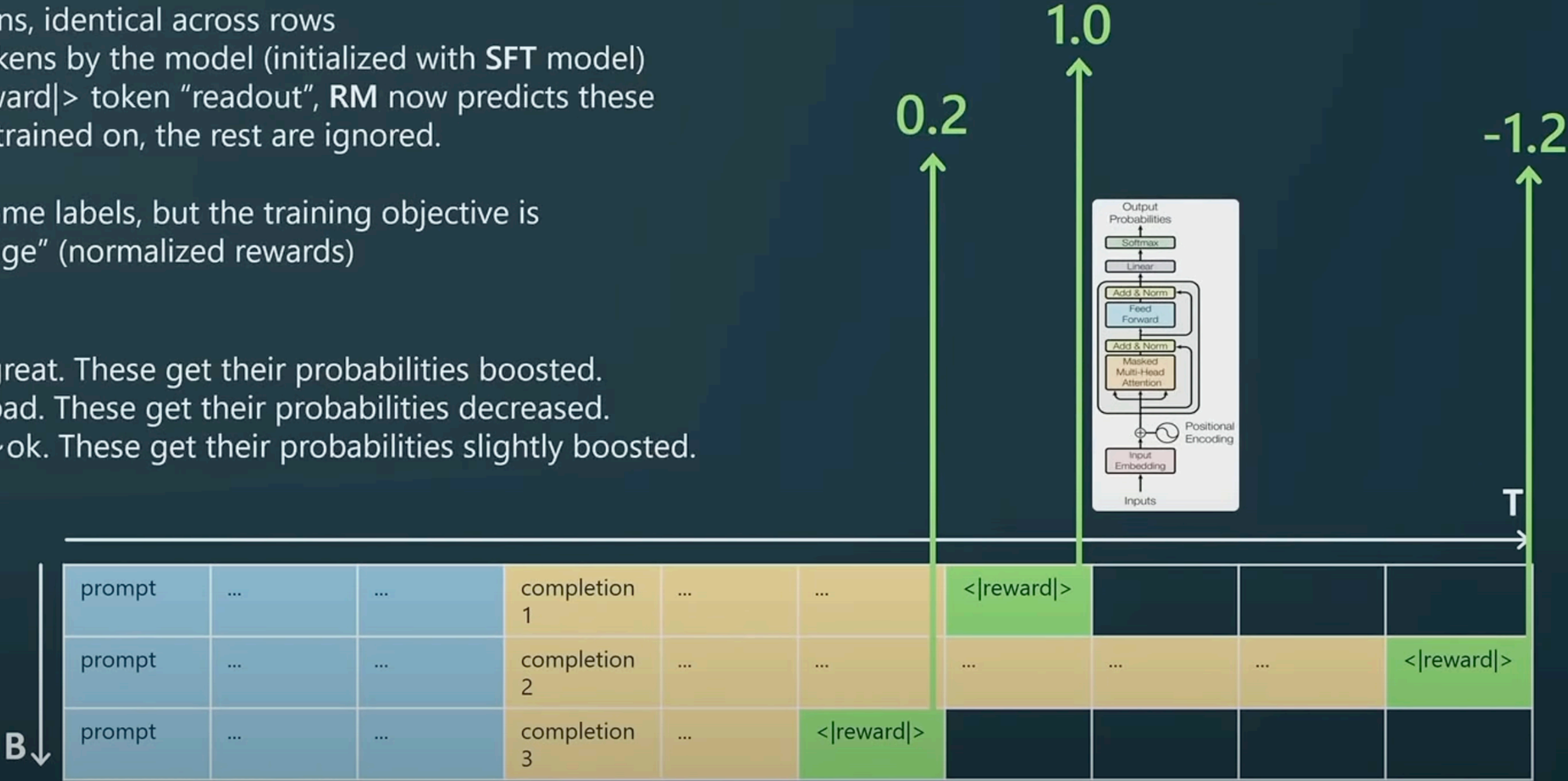
Reinforcement Learning

Blue are the prompt tokens, identical across rows
Yellow are completion tokens by the model (initialized with SFT model)
Green is the special $\langle \text{reward} \rangle$ token "readout", RM now predicts these
Only the yellow cells are trained on, the rest are ignored.

The sampled tokens become labels, but the training objective is weighted by the "advantage" (normalized rewards)

In this example:

- Row #1 tokens were great. These get their probabilities boosted.
- Row #2 tokens were bad. These get their probabilities decreased.
- Row #3 tokens were ~ok. These get their probabilities slightly boosted.



$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} \left[r_{\theta}(x, y) - \beta \log \left(\pi_{\phi}^{\text{RL}}(y \mid x) / \pi^{\text{SFT}}(y \mid x) \right) \right]$$

- Let ϕ be the parameters for the language model.
- Parameters for the <reward> token are kept frozen.
- π_{ϕ}^{RL} is the learned RL policy
- π^{SFT} is the learned supervised fine-tuning model
- β is the KL reward coefficient
- Training for chatGPT (probably) uses an actor-critic algorithm similar to proximal policy optimization (PPO) for training the ϕ parameters

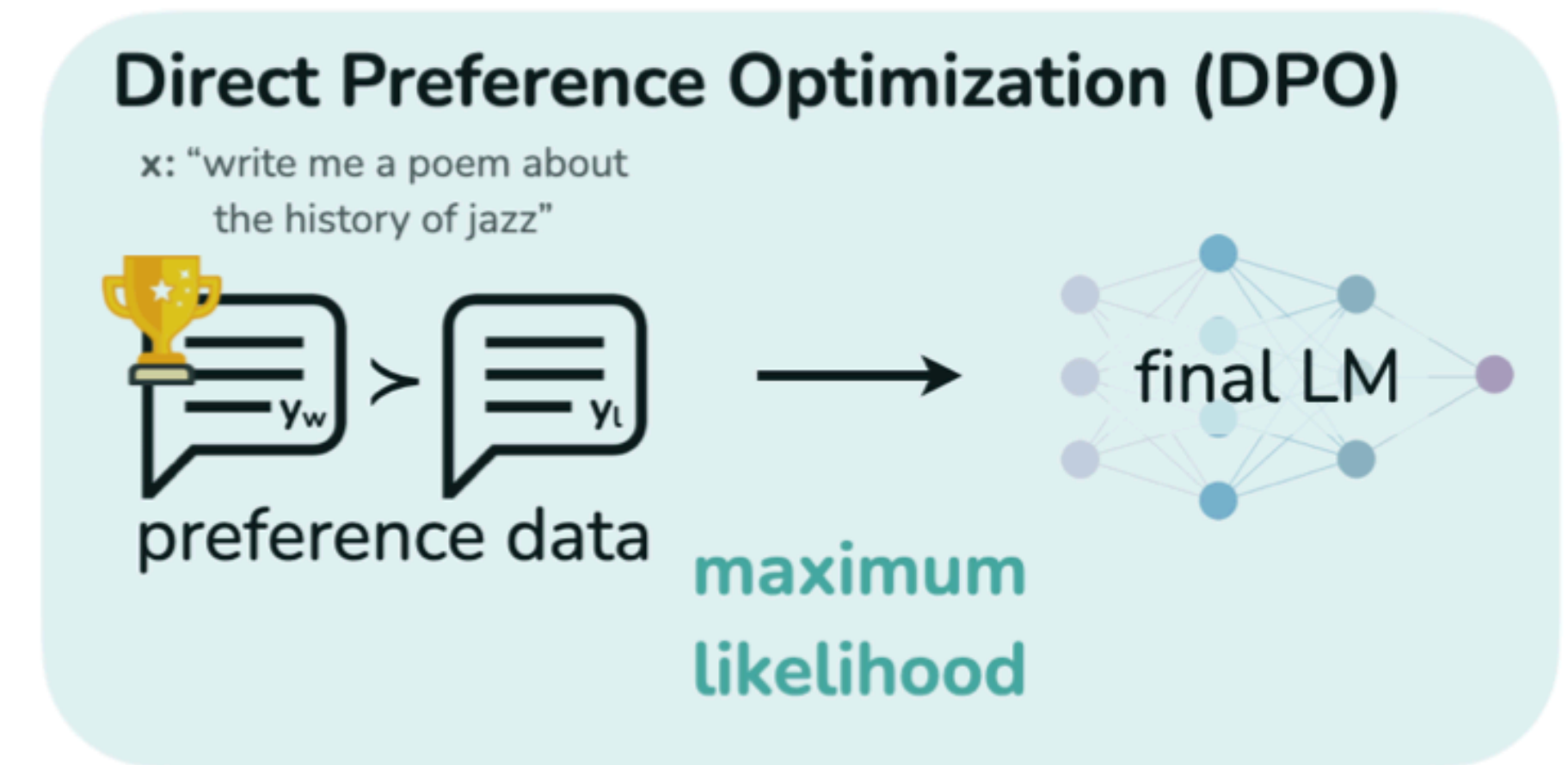
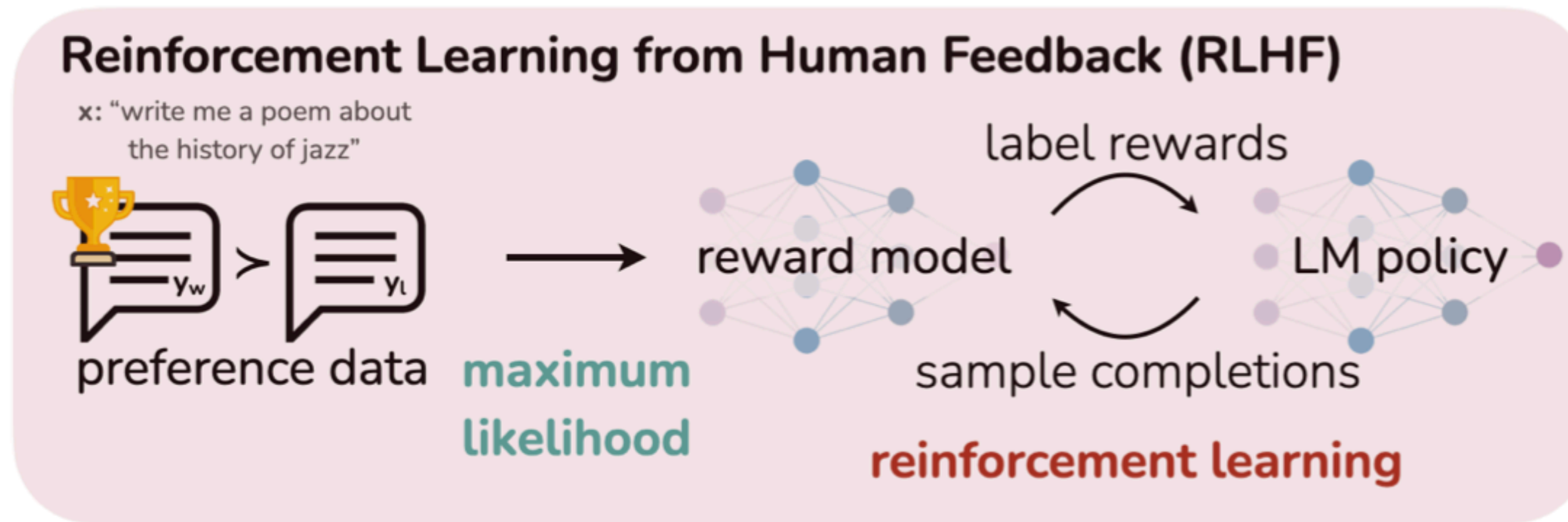
Actor-Critic RL

<https://arxiv.org/pdf/1607.07086v2.pdf>

- Standard methods to apply RL in LMs involve producing the expected reward of generating a token and generating a per-token loss for each position
- The REINFORCE algorithm is the standard way to do this for language models
- However, REINFORCE only uses a single sample token to compare against (compare y_w with y_l where $p_{y_w} > p_{y_l}$)
- Instead the actor-critic approach uses two LMs: one is the critic and one is the actor
- The critic model is trained against the reward model to produce $\langle | \text{reward} | \rangle$ at the end
- The actor model is trained against the critic and produces $\langle | \text{end of text} | \rangle$ at the end and is trained against the critic output for each time step

Direct preference optimization

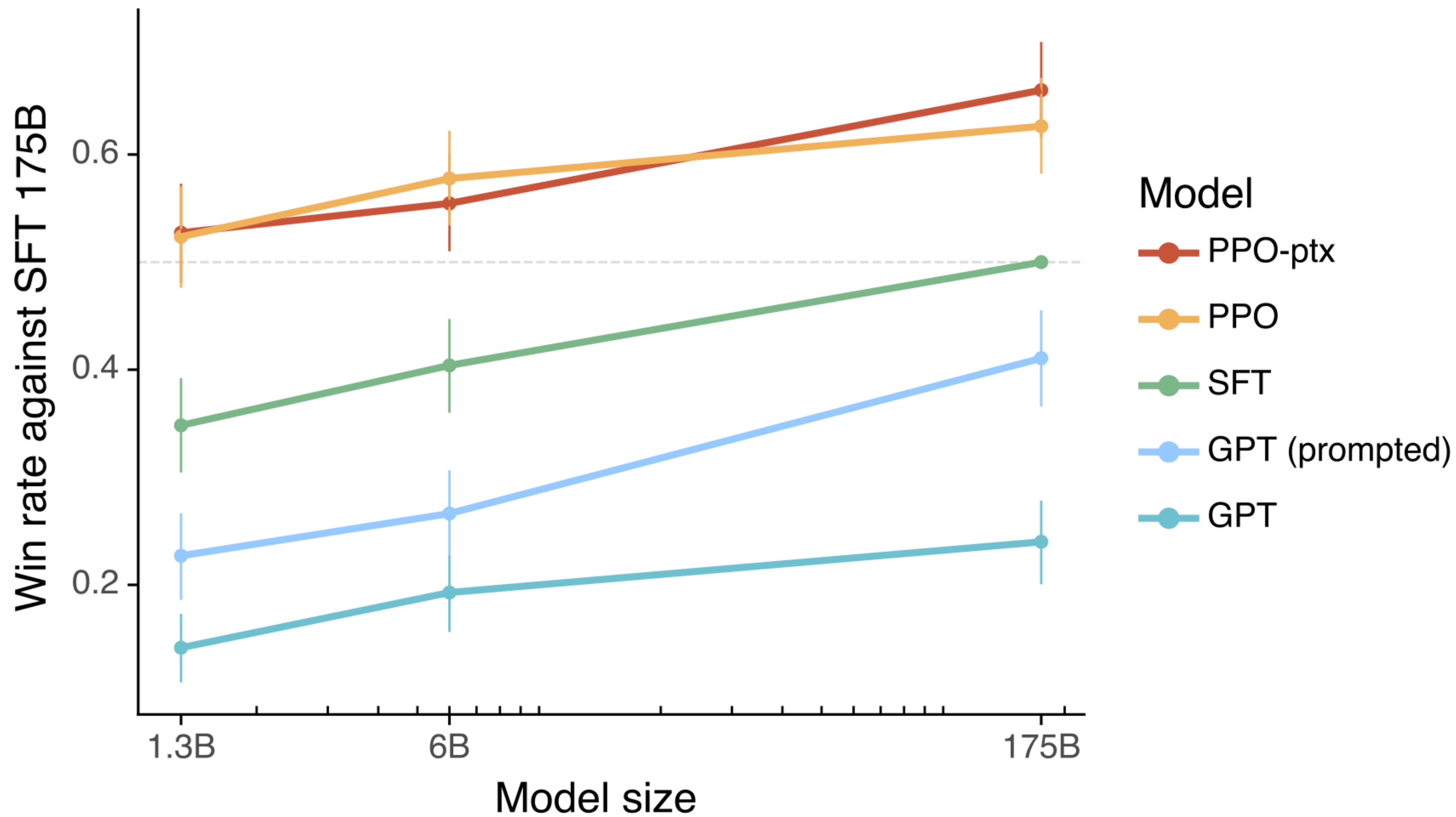
aka, Your Language Model is Secretly a Reward Model

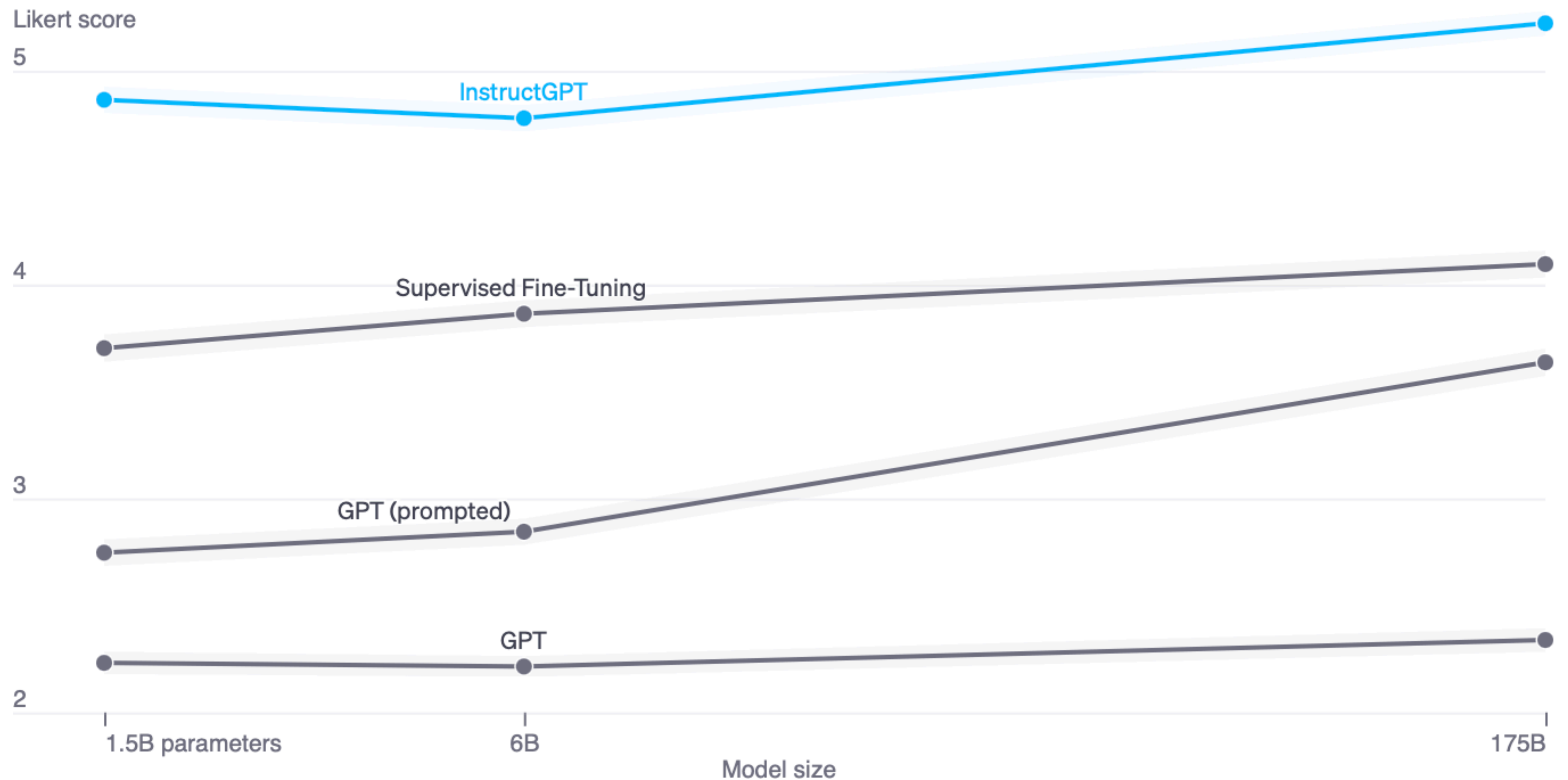


$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

$$\nabla_{\theta} \mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_{\theta}(x, y_l) - \hat{r}_{\theta}(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_{\theta} \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_{\theta} \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right]$$

Why RLHF?

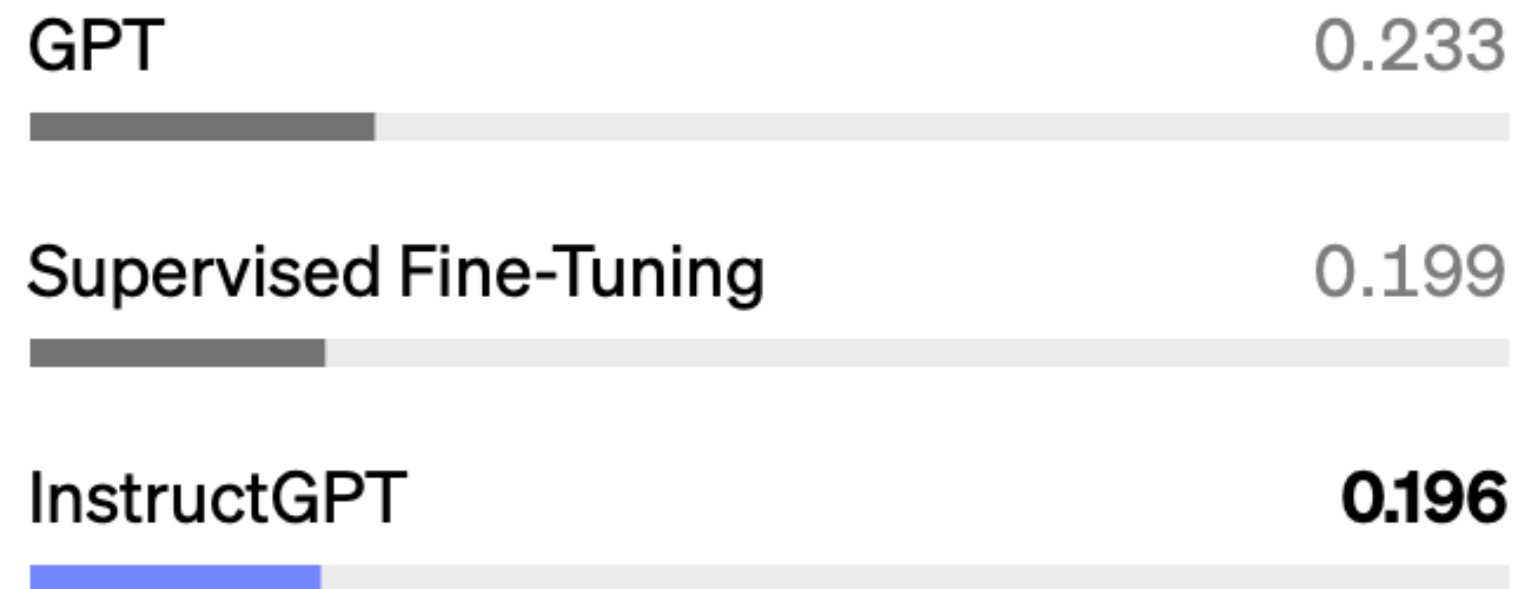




Quality ratings of model outputs on a 1–7 scale (y-axis), for various model sizes (x-axis), on prompts submitted to InstructGPT models on our API. InstructGPT outputs are given much higher scores by our labelers than outputs from GPT-3 with a few-shot prompt and without, as well as models fine-tuned with supervised learning. We find similar results for prompts submitted to GPT-3 models on the API.

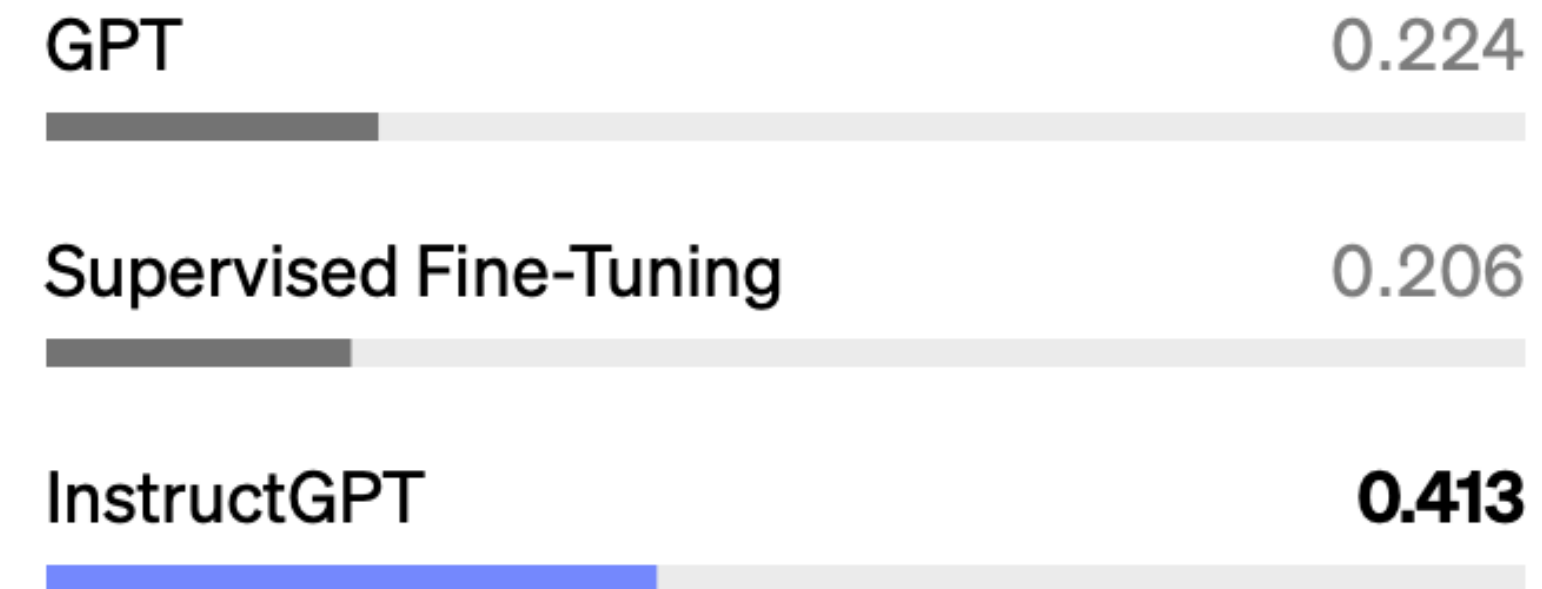
Dataset

RealToxicity



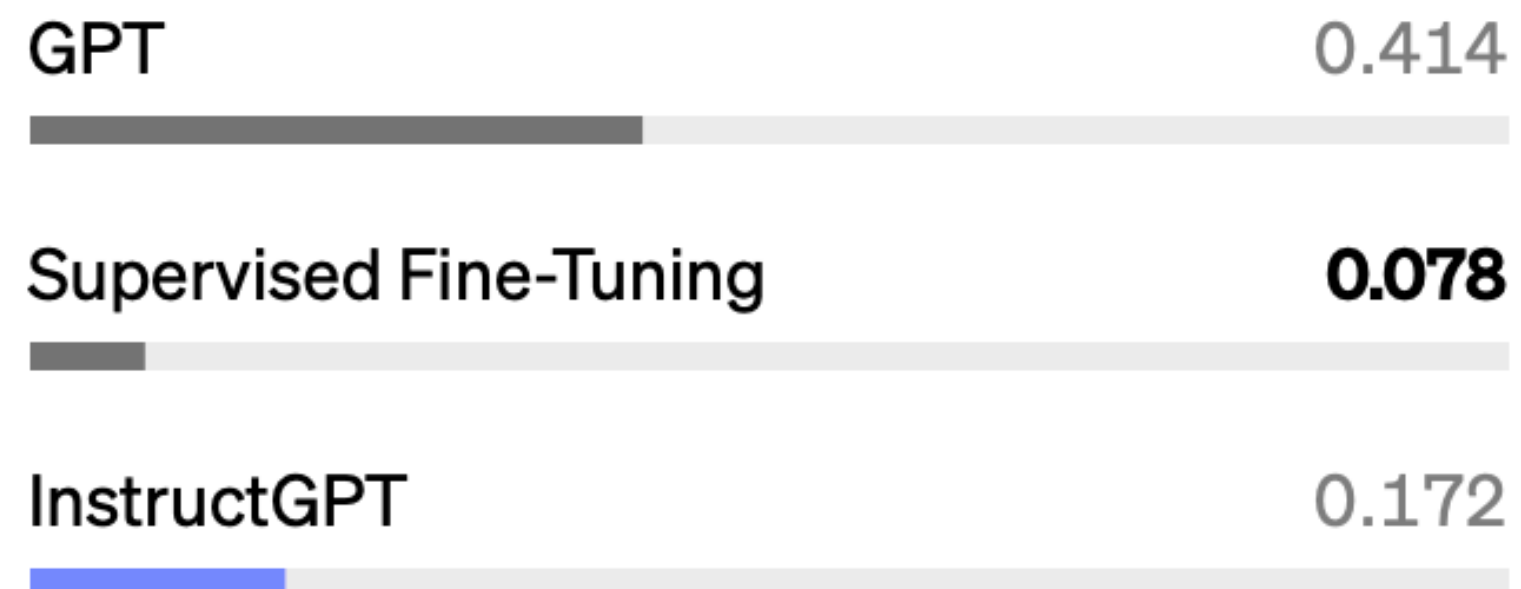
Dataset

TruthfulQA



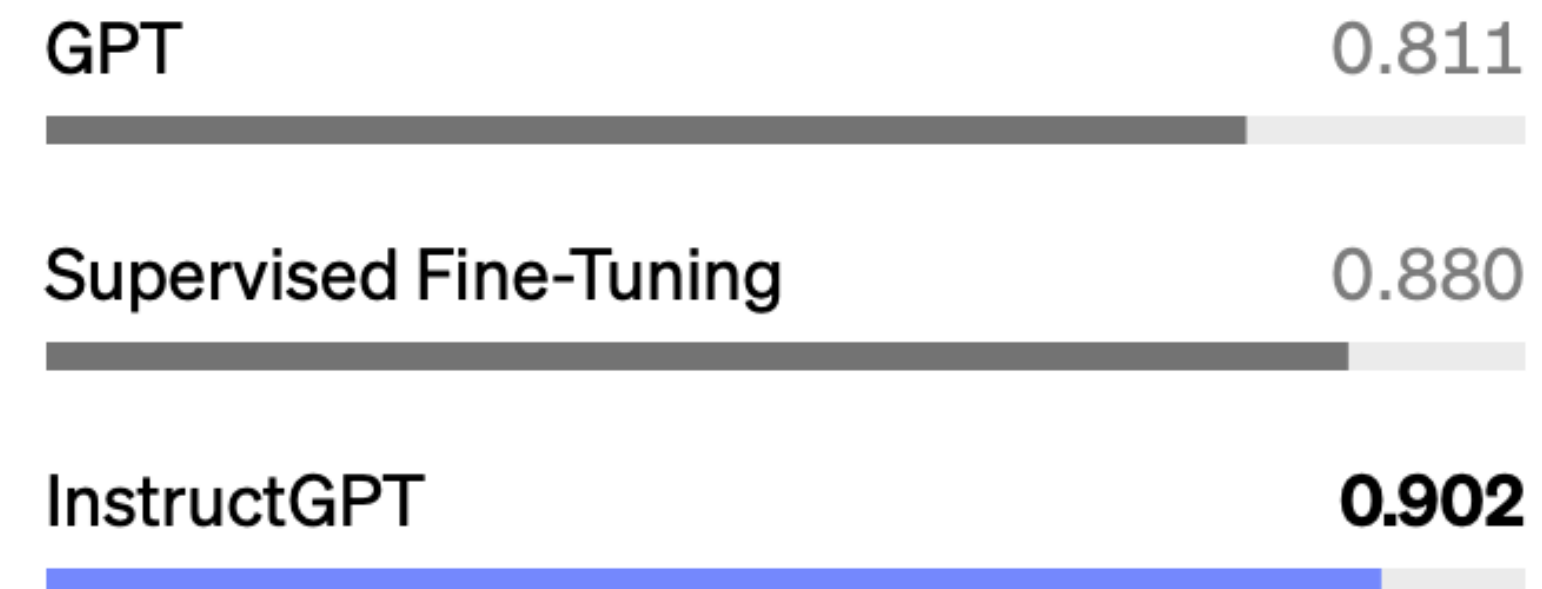
API Dataset

Hallucinations



API Dataset

Customer Assistant Appropriate



Evaluating InstructGPT for toxicity, truthfulness, and appropriateness. Lower scores are better for toxicity and hallucinations, and higher scores are better for TruthfulQA and appropriateness. Hallucinations and appropriateness are measured on our API prompt distribution. Results are combined across model sizes.

<https://openai.com/research/instruction-following>

Why RLHF?

- It is often easier to discriminate than generate
- Simple example: It is much easier to spot a bad haiku than generate one
- Writing a haiku or writing a summary or writing a story from scratch is a difficult task for humans.
- Humans are better at picking a good example by comparing to other examples.

Problems with RLHF

- Mode Collapse
- Fine-tuned models lose entropy compared to original LLM (base model)
- RLHF models confidently output very few variations
- Base models can be better at tasks that require diverse outputs

<https://www.lesswrong.com/posts/t9svvNPNmFf5Qa3TA/mysteries-of-mode-collapse>

SYSTEM

Q. Are bugs real?

A.

ASSISTANT

Yes, bugs are real.

ASSISTANT

Yes, bugs are real.

ASSISTANT

Yes, bugs are real. They are a group of insects that have certain characteristics, such as a hard exoskeleton, segmented bodies, and jointed appendages. Bugs include insects like beetles, butterflies, ants, and bees.

ASSISTANT

Yes, bugs are real. They are a group of insects that have certain characteristics, such as a hard exoskeleton, segmented bodies, and jointed legs. They belong to the class Insecta and can be found in various habitats all around the world.

ASSISTANT

Yes, bugs are real. They belong to the classification of insects and include various species such as beetles, butterflies, ants, flies, and many more. Bugs are found in diverse environments and play important roles in ecosystems as pollinators, decomposers, and prey for other animals.

ASSISTANT

Yes, bugs are real. Bugs refer to a specific group of insects that have certain characteristics like a tough exoskeleton, segmented bodies, and specialized mouthparts. They play important roles in ecosystems as pollinators, decomposers, and as a food source for other animals.

RLHF vs. Base LM

- Labelers significantly prefer InstructGPT outputs over outputs from GPT-3
- InstructGPT models show improvements in truthfulness over GPT-3 (on the Truthful QA task)
- InstructGPT shows small improvements in toxicity over GPT-3, but not bias (on the RealToxicityPrompts dataset)
- Can minimize performance regressions on public NLP datasets by modifying our RLHF fine-tuning procedure (by mixing in the pretrained distribution)

RLHF vs. Base LM

- Our models generalize to the preferences of “held-out” labelers that did not produce any training data
- Public NLP datasets are not reflective of how our language models are used
- InstructGPT models show promising generalization to instructions outside of the RLHF fine-tuning distribution
- InstructGPT still makes simple mistakes

Self-Instruct

- Generate task instructions using LLMs to train/fine-tune LLMs!

